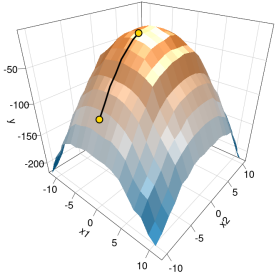
Interpretable Machine Learning

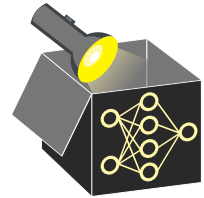
Marginal Effects



Learning goals

- Why parameter-based interpretations are not always possible for parametric models
- How marginal effects can be used in such cases
- Drawbacks of marginal effects
- Model-agnostic applicability

INTERPRETATION OF SIMPLE MODELS



- LM: Change in x_1 by Δx_1 results in change in y by $\Delta y = \Delta x_1 \cdot \theta_1$

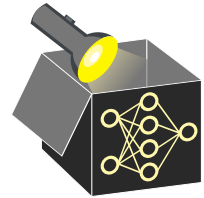
$$y = \theta_0 + \theta_1 x_1 + \cdots + \theta_p x_p + \cdots + \epsilon$$

- Default interpretations correspond to $\Delta x_1 = 1$, i.e., $\Delta y = \theta$
- All under "ceteris paribus", i.e., all remaining features are kept constant
- For polynomial models with higher-order or interaction terms, interpretations via single coefficients are not possible anymore, e.g.:

$$y = \theta_0 + \theta_1 x_1^2 + \theta_2 x_2^2 + \theta_{1,2} x_1 \cdot x_2 + \epsilon$$

  - Marginal effect of feature x_1 varies across different values for feature x_2 (and vice versa)
  - Interaction depends on values of another feature

MARGINAL EFFECTS



- Derivative ME: Derivative of f w.r.t. feature. Compute via numeric differentiation

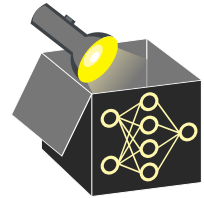
$$dME_j(x) = \frac{\partial f(x)}{\partial x_j} \approx \frac{f(x+h) - f(x-h)}{2h}$$

- Less known definition based on forward differences: Change in predicted outcome due to a change in feature value (e.g., increasing a feature x_j by h_j) $\Rightarrow$ forward ME (fME):

$$fME_j(x, h_j) = f(x_1, \ldots, x_j + h_j, \ldots, x_p) - f(x)$$

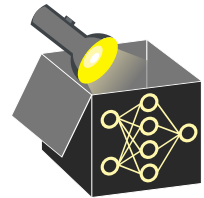
- Example: For model equation $y = \theta_0 + \theta_1 x_1^2 + \theta_2 x_2^2 + \theta_{1,2} x_1 \cdot x_2 + \epsilon$
  - dME of x_1 is $dME_1(x) = 2\theta_1 x_1 + \theta_{1,2} x_2$
  - fME of x_1 with step size $h_1 = 1$ is $fME_1(x, h_1) = 2\theta_1 h_1^2 + \theta_{1,2} x_2 h_1$

WHY MARGINAL EFFECTS *STILL* MATTER



- **Scalar summary:** $\text{FME}_j = \widehat{f}(\mathbf{x} + h\mathbf{e}_j) - \widehat{f}(\mathbf{x})$ - interpretable, testable, storable.
- **Dimension-agnostic:** Effective with many features, incl. categorical; avoids PD/ICE's dimensionality constraints.
- **Model-faithful:** Local to $\mathbf{x}$; preserves interactions without assumptions.
- **Efficient:** Two predictions per feature suffice.
- **Conclusion:**
  - ME yield concise, rigorous insights
  - Complementary to PD/ICE visualizations

USE-CASE: SCALAR VS. VISUAL ESTIMATION



Clinician's questions

- *"What if this patient's systolic BP increases by 10 mmHg?"*
- *"What if BP increases by 10 mmHg & LDL by 15 mg/dL?"*

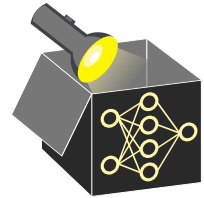
Route A – ICE / PD

- Visualize effect along 1-D or 2-D feature grid.
- Manual interpretation of curve/surface deltas.
- $\rightarrow$ local, visual, interaction-limited.

Route B – Forward Marginal Effect: $\mathrm{FME}_{\Delta\mathbf{x}} = \widehat{f}(\mathbf{x} + h) - \widehat{f}(\mathbf{x})$

- **1-D case:** $\Delta\mathbf{x} = (10, 0) \;\Rightarrow\; +\mathbf{3}$ pp
- **2-D case:** $\Delta\mathbf{x} = (10, 15) \;\Rightarrow\; +\mathbf{4.1}$ pp
- One scalar answer per query, extensible to higher dimensions.

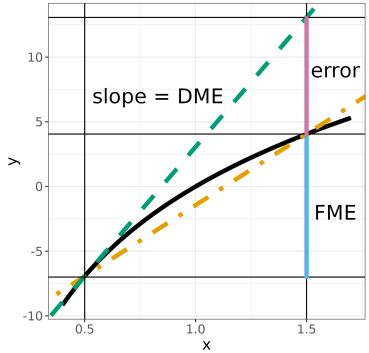
RELATION TO ICE AND PD

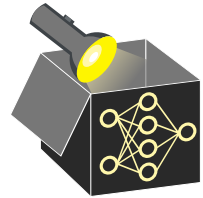


- **Individual Conditional Expectation (ICE)**:
  - Visualizes predictions for an observation across a range of feature values.
  - FME corresponds to vertical differences between points on an ICE curve.
- **Partial Dependence (PD)**:
  - Shows average predictions across a range of feature values.
  - AME is equivalent to vertical differences on PD for linear models.
- **Advantages of FMEs**:
  - Provide exact change in prediction.
  - Applicable to high-dimensional feature changes.
  - Quantifiable and not limited to visual interpretation.

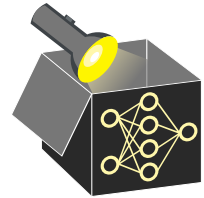
DERIVATIVE VERSUS FORWARD DIFFERENCE

- Derivative of non-linear prediction functions substantially differ at different points
  $\Rightarrow$ dME is not suited
- fME corresponds to a movement on prediction function, indicating changes in predicted outcome regardless of the function's shape
  $\Rightarrow$ fME better suited for non-linear prediction functions
- However, with both variants, we lose information about the prediction function along the finite difference





ADDITIVE RECOVERY

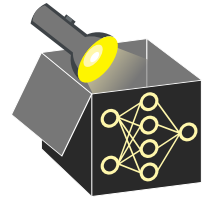


- Both variants only recover terms within prediction function that depend on feature(s) of interest
- Consider a prediction function $\widehat{f}(x) = ax_1 + bx_2$. It follows that:

$$dME_1(x) = a$$
$$fME_1(x, h_1) = ah_1$$

- MEs remove effects of other features that are linked additively

MODEL-AGNOSTIC APPLICABILITY



- MEs were historically used to interpret GLMs, however, they can be used as model-agnostic interpretation tools for non-parametric models
- fMEs correspond to an exploration of the prediction function and are better suited for non-linear prediction functions
- With fMEs, we can also use multivariate changes in feature values to explore the prediction surface in various directions simultaneously
- Note: Shape of the prediction function may vary considerably along the forward difference, i.e., an fME with half the step size may not result in half the fME